

SNAPFRAME · DESIGN DECISIONS REGISTER

SnapFrame design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Amish accepted the cost overrun against the value-engineering target on 2026-10-03; row added to decisions made; value engineering section updated
0.3	2026-10-02	Amish Chadha	Decisions carried into the design: value engineering restated at \$700.56 (nodes in filled nylon, folding step, long anchors, straps); two new points found while doing so, proposed and awaiting Amish (node print cost, 1 in posts); To confirm items updated
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all ten open decisions (SNF-DDR-003 accepted); moved to decisions made; To confirm items 4 and 6 and the value engineering note updated
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items from SNF-DDR-001 to SNF-DDR-003 and the review note; budget treated as a value-engineering target

SnapFrame design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Open decision 11 (cost) was decided on 2026-10-03. One new point found on 2026-10-02 while carrying the decisions of that day into the design is still open:

#	Decision	Options	Recommendation	What it affects in the build	Source
12	Posts: the frame analysis is run (SNF-CAL-001 v0.6) and the windward 3/4 in post governs at a factor of 1.43 at 20 m/s (rating 19.5 m/s; 20.3 m/s with 1 in posts, about \$30 more). The decision of 2026-10-02 fits 1 in posts only if the analysis had not been run, so the posts stay 3/4 in.	(a) keep 3/4 in posts and rate the kit at 19.5 m/s, as decided; (b) fit 1 in posts (6 sticks at \$14.00 instead of \$9.00, the 12 post sockets of the foot and eave nodes at 30.1 mm) and rate it at 20.3 m/s; (c) keep 3/4 in and test the knee fixity at TRL 4, which would give 20.0 m/s at best	(a) now, with (c) at TRL 4 and (b) held in reserve	Posts, foot nodes and eave nodes (model, BOM lines 1, 5, 6)	SNF-DDR-002 O8; SNF-CAL-001 v0.6 section 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Snap buttons for 3/4 in and 1 in EMT: the spring reaches no more than 45 mm deeper than the button, and the button stands at least 3.5 mm above the tube surface	The spring must clear the hitch pin 15 mm from the tube end, and the button must stand proud in its finger recess	SNF-DDR-003, C1 and C2
2	Snap hooks: the gate opens at least 14 mm, to pass the 12 mm wire of the anchor eye, and they fit the 6 mm ring	Every cable end clips to an anchor eye or a ring	SNF-DDR-003, C3 and C4
3	Welded 6 mm ring of 32 mm bore: working load at least 2 kN; hand tensioner for 4 mm wire: working load at least 1.5 kN	The highest cable tension is 1.21 kN at 20 m/s; the tensioner rating has been unknown since TRL 3	SNF-CAL-001, section 6
4	Screw anchor: eye wire about 12 mm and eye about 56 mm across, so it straddles the 18 mm slot; holding capacity in the soils of the first site; a longer anchor of the same eye for the two rear corner feet	The eye clamps the foot plate; R8 needs 1.5 kN at the two rear corner feet and 1.0 kN at the others (decided 2026-10-02)	SNF-DDR-003, C5; R8
5	EMT yield strength and dimensions from the supplier	The wind rating assumes 275 MPa yield	SNF-CAL-001, A3
6	Print farm bed at least 230 mm, and that it can print glass- or carbon-filled PA12-class nylon (decided 2026-10-02), with a quote per kilogram	The eave and middle ridge nodes are 220 mm across; the node prices assume \$55/kg and \$1.50 a node	SNF-DDR-003; decision of 2026-10-02 on the node polymer; SNF-CAL-001 A12
7	Folding step: top tread 0.50 m, about 1.5 kg, rated for 100 kg or more, folded 0.50 x 0.45 x 0.08 m	A person standing on the floor reaches about 2.2 m and the ridge sockets are at 2.6 m	SNF-DDR-003 A1 (decided 2026-10-02); BOM line 15
8	Long screw anchor, 560 mm, rated 1.5 kN in firm soil, same 12 mm wire eye as the standard anchor	The two rear corner feet carry up to 1.28 kN	SNF-CAL-001 section 6; R8; BOM line 16

■ Value engineering

Value-engineering target: USD 445. Estimated cost of the constructable design: USD 700.56 (USD 255.56 over the target). This is the size M frame kit; the target is a hypothetical control target, not a limit. Tarpaulins are agency stock and costed separately (three, USD 75.00). The estimate rose from USD 469.14 (SNF-CAL-001 v0.5) because of decisions made on 2026-10-02. Main cost drivers and savings worth trying:

Amish accepted this overrun on 2026-10-03: the estimated cost of USD 700.56 against the USD 445 target (USD 255.56 over), with the nodes in filled nylon (open decision 11, option a). Amish: "Cost over target - i accept all the cost variations and overruns". It stays reported against the target as an accepted overrun, and the savings below remain worth trying.

- The largest lines are the 15 printed nodes (USD 316.16 in filled PA12-class nylon, of which the six foot nodes are USD 160.86), the tubes (USD 202.00 for 18 sticks, 39 % of the tube bought ends up as offcut), the brace cables (USD 42.00), the anchors (USD 36.00, six standard and two long), the folding step (USD 25.00), the buttons, pins and caps (USD 24.00) and the cable bolt sets (USD 23.40).

- What changed on 2026-10-02: the nodes rose by USD 196.42 (filled nylon at USD 55/kg and 1,200 kg/m³ instead of ASA at USD 22/kg and 1,070 kg/m³, and USD 1.50 of machine time a node instead of USD 1.00); the folding step added USD 25.00; the two long anchors added USD 4.00 net of the two standard anchors they replace; the second set of straps added USD 6.00.
- Savings worth trying: a print farm quote for the filled nylon (open decision 11, accepted as an overrun on 2026-10-03, so the quote is optional); lower infill on the low-stress ridge nodes once their coupons are tested; a lighter foot plate, since the foot node is the most expensive print (USD 26.81 each); bulk or 6 m metric tube, which also cuts the offcut (the 39 % offcut is accepted for the prototype, decided 2026-10-02); casting the nodes in aluminum later; and stainless hardware bought by the hundred for the cable bolt sets.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: budget covers the frame kit, tarpaulins from agency stock; 1 in EMT rafters; print nodes, design for casting; spring buttons plus hitch pins at the 12 tension joints; gable with cable bracing; sizes S, M and L with M as reference; snow and cyclone out of scope	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	SNF-DDR-001
2026-09-25	D8: 1 in EMT ridge tubes, frame analysis before any decision on posts. D9: one four-socket eave node at all six positions, the corner socket past the gable left blank and capped	Amish: "i accept all your recommendations, go with them across all repos."	SNF-DDR-002
2026-09-26	Budget set to \$445 to cover the priced frame kit (O4 closed)	Amish: "i approve all the budget items."	SNF-DDR-002 v0.2
2026-09-30	Prototype build plan in the approved format; outstanding decisions kept out of the build plan and in this register	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"; "don't log outstanding decisions in this build plan - that is not the place for it"	.kit/STANDARDS.md section 18
2026-09-30	Fix the design where it cannot be built as drawn, keeping what it does	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SNF-DDR-003 (the changes themselves are open decision 1)
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit; cost reported as over or under it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	.kit/STANDARDS.md section 18
2026-10-02	Design for construction accepted: the nine changes C1 to C9 as made (open item 1)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-003, C1 to C9
2026-10-02	Ridge tubes: the first prototype uses a folding step; guiding from the ground with a spare tube (option b) is timed in the TRL 4 user trial, and the step stays in the kit if ground guiding is slower or needs a third person (open item 2)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-003, A1
2026-10-02	Front gable closed with part of a third tarpaulin from agency stock, cut to include a door flap; the corner cable hem is settled in the same cutting plan; R7 restated to match (open item 3)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-001, O3; SNF-DDR-003, A2; SNF-REQ-001 R7
2026-10-02	Posts: the frame analysis is run now, as paper work at TRL 3; if it has not been run when the		SNF-DDR-002, O8

Date	Decision	Decided by	Record
	prototype tubes are bought, 1 in posts are fitted (open item 4)	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-02	Packages: the tubes are split into two bundles (16.5 and 12.1 kg) and R5 restated as three packages of 25 kg or less; R11 is counted on the frame kit alone (43.3 kg against 50 kg), since the tarpaulins are issued separately from agency stock (open item 5)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-002, 05; SNF-DDR-003
2026-10-02	Node polymer: a glass- or carbon-filled nylon of the PA12 class, with printed coupons tested for pin bearing at 70 °C at TRL 4; PETG dropped (open item 6)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-002, 06
2026-10-02	Anchors: the target at the two rear corner feet is raised to 1.5 kN, with longer anchors there; 1.0 kN is kept at the other feet (open item 7)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-002, 07
2026-10-02	Tube offcut of 39 % accepted for the prototype on US 10 ft stock; the choice between 1.50 m posts and 6 m metric stock is made once the first region's tube source is known (open item 8)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-001, 01
2026-10-02	Partners: the IFRC Shelter Research Unit is the first candidate to approach for a technical review of the kit with standard relief tarpaulins, and Field Ready as the printing partner; the region is taken from the first warm-climate response the review points to (open item 9)	Amish: "i approve your recommendations for all 555 open decisions."	SNF-DDR-001, 02
2026-10-02	Renderers: the first three render choices accepted (tube split at the detail node, hero without guy lines, foot plate grip ribs and size mark as appearance only); the fourth (brace cable start at the detail node) withdrawn when the renders are redone (open item 10)	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26 render session, items 1 to 4
2026-10-03	Cost overrun accepted: the estimated cost of USD 700.56 against the USD 445 target (USD 255.56 over), with the nodes in filled nylon (open decision 11, option a)	Amish: "Cost over target - i accept all the cost variations and overruns"	REVIEW.md, session 2026-10-03